

* Factorial (! or L)

Let $n \in \mathbb{N}$, then factorial of n is denoted by $n!$ or $!n$ (read as n factorial or factorial n) and is defined as the product of all integers from n to 1.

E.g. ①: $5! = 5 \times 4 \times 3 \times 2 \times 1$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$\textcircled{2}: n! = n(n-1)!$$

$$n! = n(n-1)(n-2)!$$

$$n! = n(n-1)(n-2) \dots \times 1$$

Note: ① $0! = 1$

Proof:

We know

$$n! = n(n-1)!$$

$$(n-1)! = \frac{n!}{n}$$

When $n=1$

$$(1-1)! = \frac{1!}{1}$$

$$0! = \frac{1}{1}$$

$$0! = 1$$

② Factorial of negative integer is not defined

Permutation

→ It is an arrangement of objects in definite order by taking some or all at a time. The number of permutations of n objects taken r at a time is denoted by $P(n, r)$ or ${}^n P_r$.

e.g: @ $P(5, 3) = {}^5 P_3 = 60$

Theorem 1: The total number of permutation of n different objects taken r at a time is given by,

$$P(n, r) = \frac{n!}{(n-r)!} \quad ; (r \leq n)$$

Proof

The total number of permutations of n different objects taken r at a time is equivalent with the ways of filling of r places by using n different object.

So

1st place has n ways

2nd place has $(n-1)$ ways

3rd place has $(n-2)$ ways

4th place has $(n-3)$ ways

⋮

r^{th} place has $(n - (r-1))$ ways

$= (n - r + 1)$ ways

∴ Total no. of permutation = $n(n-1)(n-2)(n-3) \dots (n-r+1)$

$$\text{or, } P(n, r) = \frac{n(n-1)(n-2) \dots (n-r+1) [(n-r) \cdot (n-r-1) \dots 1]}{[(n-r) \cdot (n-r-1) \dots 1]}$$

$$\Rightarrow P(n, r) = \frac{n!}{(n-r)!}$$

• Corollary

① When $r=n$, we get

$$P(n, n) = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$$

② When $r=(n-1)$, we get

$$P(n, n-1) = \frac{n!}{(n-n+1)!} = \frac{n!}{1!} = n!$$

Which is maximum number of permutation.

Theorem 2: Total no. of permutations of n objects taken all at a time out of which p are of 1st kind, q are of 2nd kind and r are of 3rd kind and rest are different, is

$$\frac{n!}{p!q!r!}$$

E.g. ① In how many ways can you arrange the letters of the words:

• MATHEMATICS

→ Soln,

Total no. of letters (n) = 11

M occurs two times (p) = 2

A occurs two times (q) = 2

T occurs two times (r) = 2

∴ Total no. of permutation = $\frac{n!}{p!q!r!}$

$$= \frac{11!}{2!2!2!}$$

$$= \frac{11!}{2!2!2!}$$

• ELEMENT

→ Soln,

Total no. of letters $(n) = 7$

E occurs three times $(P) = 3$

∴ Total no. of permutations = $\frac{n!}{P!}$

$$= \frac{7!}{3!} = 840$$

• EDUCATION

→ Soln,

Total no. of letters $(n) = 9$

All different letters so,

Total no. of permutations = $9! = 362880$

• APPLE

→ Soln,

$n = 5$

P occurs 2 times $(P) = 2$

Total no. of permutations = $\frac{5!}{2!} = 120$

Theorem 3 : Total number of permutation of n objects taken r at a time if the repetition is allowed is n^r .

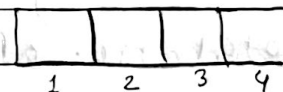
E.g. ① In How many can a person post 3 letters in 4 boxes?

→ Soln,

1st letter has 4 ways

2nd letter has 4 ways

3rd letter has 4 ways



$$4 \times 4 \times 4$$

$$\begin{aligned}\text{Required no. of ways} &= 4 \times 4 \times 4 \\ &= 4^3 \\ &= 64\end{aligned}$$

Exercise 1.2

part B

① a) Solve for n , if $\frac{(n+1)!}{(n-1)!} = 12$ Where n is whole number.

↳ Soln,

$$\text{Given: } \frac{(n+1)!}{(n-1)!} = 12$$

$$\Rightarrow \frac{(n+1) \cdot n \cdot (n-1)!}{(n-1)!} = 12$$

$$\Rightarrow \frac{(n+1) \cdot n \cdot (n-1)!}{(n-1)!} = 12$$

$$\Rightarrow n^2 + n - 12 = 0$$

$$\Rightarrow n^2 + 4n - 3n - 12 = 0$$

$$\Rightarrow n(n+4) - 3(n+4) = 0$$

$$\Rightarrow (n-3)(n+4) = 0$$

Either,

or,

$$n-3=0$$

$$n+4=0$$

$$n=3$$

$$n=-4 \text{ (Not whole number)}$$

Since n is a whole number

$$\therefore n=3$$

b) If $P(5, r) = 5$, find the value of r :

↳ Soln,

$$P(5, r) = 5$$

$$\Rightarrow \frac{5!}{(5-r)!} = 5$$

$$\Rightarrow \frac{5!}{5} = (5-r)!$$

$$\Rightarrow \frac{5 \times 4!}{5} = (5-r)!$$

$$\Rightarrow 4! = (5-r)!$$

$$\Rightarrow (5-1)! = (5-r)!$$

$$\therefore r=1 \text{ (Equating L.H.S \& R.H.S)}$$

c) Show that: $(2n)! = (1 \cdot 3 \cdot 5 \cdots (2n-1)) \cdot 2^n n!$

→ Soln,

$$\text{L.H.S} = (2n)!$$

$$= 2n (2n-1) (2n-2) (2n-3) \cdots 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= [(2n-1) (2n-3) \cdots 5 \cdot 3 \cdot 1] \cdot [2n (2n-2) (2n-4) \cdots 4 \cdot 2]$$

$$= [1 \cdot 3 \cdot 5 \cdots (2n-1)] [2 \times 2 \times 2 \times 2 \cdots \times 2] [n(n-1)(n-2) \cdots 2 \cdot 1]$$

$$= [1 \cdot 3 \cdot 5 \cdots (2n-1)] 2^n n!$$

$$= \text{R.H.S}$$

Proved

② In how many ways can 4 boys and 3 girls be seated in a row containing 7 seats if they sit anywhere? If they seat alternately how many such arrangements are possible?

→ Soln,

1st part (if they sit anywhere).

- 7 persons (4B + 3G) in 7 different seats can be arranged by $7!$ ways
 $= 5040$

2nd part (they sit alternately)

- 4 boys in 4 seats can be arranged by $4!$ ways.
- 3 girls in 3 seats can be arranged

by $3!$ ways.

$$\therefore \text{Required no. of arrangements} = 4! \times 3! = 144$$

5) How many Arrangements are possible using the letters of the word EQUATION only once? If all vowels come together, how many such arrangements are possible?

→ Soln.

Given "EQUATION"

no. of letters $(n) = 8$

Total no. of arrangements = $8!$
 $= 40320$

2nd part (all vowels come together).

- Individual arrangements: $\boxed{EVAIO}$, $\underset{1}{Q}$, $\underset{2}{T}$, $\underset{3}{N}$

$\therefore$ no. of arrangements = $4!$

- Block Arrangements : E, U, A, I, O
1 2 3 4 5

$\therefore$ no. of arrangements = 5!

So,

Required no. of arrangements = $4! \times 5!$
 $= 2880$

3) How many 6 digits numbers can be formed by using the digits 0 to 9 only once? How many such numbers are divisible by 10?

↳ Soln,

For 6 digits numbers

First digit shouldn't contain 0 because it makes 5 digits number:

So,

First digit has 9 choice

Second digit has 9 choice

Third digit has 8 choice

Fourth digit has 7 choice

Fifth digit has 6 choice

Sixth digit has 5 choice

6 5 4 3 2 1

$9 \times 9 \times 8 \times 7 \times 6 \times 5$

$$\therefore \text{Required no. of digits} = 9 \times 9 \times 8 \times 7 \times 6 \times 5 \\ = 136080$$

2nd part

For a number to be divisible by 10 must contain 0 in last digit.

So,

last digit has 1 choice

first digit has 9 choice

Second digit has 8 choice

Third digit has 7 choice

Fourth digit has 6 choice

Fifth digit has 5 choice

6 5 4 3 2 1

$5 \times 6 \times 7 \times 8 \times 9 \times 1$

$$\therefore \text{Required no. of digits} = 9 \times 8 \times 7 \times 6 \times 5 \times 1 \\ = 15120$$

- 4) How many numbers lying between 100 and 1000 can be formed with the digits 0, 1, 2, 3, 4, 5 if the repetition of the digits is not allowed?

→ Soln,

Total no. of digits $(n) = \{0, 1, 2, 3, 4, 5\} = 6$

no. of place $(r) = 3$

Now,

No. of ways = $\frac{n!}{(n-r)!}$
(including start
With 0)

$$= \frac{6!}{(6-3)!}$$

$$= \frac{6!}{3!}$$

$$= \frac{6 \times 5 \times 4 \times 3!}{3!}$$

$$= 120$$

No. of invalid Numbers [starts with Zero (0)]

Ones place can be filled by 4 ways	H	T	O
Tens place can be filled by 5 ways	1	5	4
Hundreds place can be filled by 1 way	$= 20$		

∴ No. of invalid Numbers = $1 \times 5 \times 4 = 20$.

∴ Reduced No. of Valid ways = $120 - 20$

$= 100$ Ways

⑥ In how many ways can the letters of the word HOLIDAY be arranged? Also, find the number.

→ Given: HOLIDAY

Total no. of letters (n) = 7

Total no. of arrangements = $7!$
 $= 5040$

as all vowels come together.

4 Individual arrangements: $\boxed{O C A}$, H, L, D, Y
1 2 3 4 5

no. of individual arrangements = $5!$

Block arrangements: O, I, A

1 2 3

no. of block arr = 3!

$$\therefore \text{Required no. of arr.} = 5! \times 3! = 720$$

b) No two vowels are together.

⑤ H ⑥ L ⑦ D ⑧ Y ⑨

3 vowels in 5 place can be arranged in $P(5,3)$ ways.

4 consonants in 4 places can be arranged in $4!$ ways.

$$\therefore \text{Required no. of arrangement} = P(5,3) \times 4! \\ = 1440$$

c) The vowels may occupy only odd places.

① ② ③ ④ ⑤ ⑥ ⑦

3 vowels in 4 place can be arranged in $P(4,3)$ ways.

4 consonants in 4 remaining place can be arranged in $P(4,4)$ ways.

$$\therefore \text{Required no. of arrangements} = P(4,3) \times P(4,4) = 576$$

d) the relative positions of vowels and consonants do not change.

3 vowels in 3 places = $P(3,3) = 3!$

4 consonants in 4 places = $P(4,4) = 4!$

$$\text{Required no. of arrangements} = 3! \times 4! = 144$$

3. In how many ways can letters of following words be arranged without any restriction?

③ EXAMINATION

Soln

Total no. of letters $(n) = 11$

'A' occurs 2 times $(P) = 2$

'I' occurs 2 times $(q) = 2$

'N' occurs 2 times $(r) = 2$

$$\begin{aligned}
 \text{required no. of arrangement} &= \frac{n!}{p! \cdot q! \cdot r!} \\
 &= \frac{11!}{2! \cdot 2! \cdot 2!} \\
 &= 4989600
 \end{aligned}$$

a) PERMUTATION

Soln

Total no. of letters (n) = 11

'T' occurs 2 times (P) = 2

$$\begin{aligned}
 \text{required no. of arrangement} &= \frac{n!}{p!} \\
 &= \frac{11!}{2!}
 \end{aligned}$$

$$= 19958400$$

b) INTERMEDIATE

Soln

Total no. of letters (n) = 12

'I' occurs 2 times (P) = 2

'T' occurs 2 times (q) = 2

'E' occurs 3 times (r) = 3

$$\begin{aligned}
 \text{required no. of arrangement} &= \frac{n!}{p! \cdot q! \cdot r!} \\
 &= \frac{12!}{2! \cdot 2! \cdot 3!}
 \end{aligned}$$

$$= 19958400$$

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10) How many sin digit numbers can be formed by using the digits 0, 2, 2, 2, 5, 5?

Soln Given: 0, 2, 2, 2, 5, 5

To form 6 digits number, first place must be either 2 or 5.

Arrangements starting with 2.

2 { 0, 2, 2, 5, 5 }

fin

no. of 6-digits numbers starting with

$$2 = \frac{P(5,5)}{2! 2!} = \frac{5!}{2! 2!} = 30$$

Also,

Arrangements starting with 5.

5 { 0, 2, 2, 2, 5 }

fin

no. of 6-digits numbers starting with

$$5 = \frac{P(5,5)}{3! 1!} = \frac{5!}{3!} = 20$$

So,

required no. of arrangements = $30 + 20$

$$= 50$$

✓

(12) In how many ways can the letters of the word ARRANGE be arranged?

How many such arrangements are having two R's together? How many of them are not having two R's together.

Soln.

Given: ARRANGE

Total no. of letters (n) = 7

'R' occurs 2 times (p) = 2

'A' occurs 2 times (q) = 2

required no. of Arrangement = $\frac{n!}{p! q!}$

$$= \frac{7!}{2! 2!}$$

$$= \frac{7!}{2! 2!}$$

Again,

Having 2 R's together.

- $$\text{no. of individual Arrangement} = \frac{6!}{2!} = 360$$

- $$\text{no. of block Arrangement} = \frac{2!}{2!}$$

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$\therefore$ Required no. of Arrangement = 360×1

$$= 360$$

~~2. 7. 20~~

And,

Not having 2P's together - $1260 - 360$
 $= 900$

∴ No. of ways of arrangement for not having 2 R's together = 900.

- ⑦ In how many ways can the letter of the word SUNDAY be arranged? How many of these arrangements do not begin with s? How many begin with s and donot end with y?

Soln.

Given: SUNDAY

no. of letter (n) : 6

required no. of arrangement = $6!$
 $= 720$

- For Arrangement begin with S.

S {U, N, D, A, Y}

fix

no. of Arrangement = $5!$

(begin with S) = 120

∴ No. of Arrangement doesn't begin

with S = $720 - 120$

= 600

- Begins With S and ends ~~ends~~ With Y.

S {U, N, D, A, Y}

fix

fix

no. of arrangement Begins With S

and ends with Y = $4!$

= 24

∴ No. of arrangement begins with S

and don't ends with Y = $120 - 24$

= 96

⑧ In how many ways can 10 boys be arranged in a row having 4 seats if

a) One particular boy is always included?

Ans. No. of ways of particular boy to be

Seated = 4

Now, remaining seat = 3

No. of ways of remaining seat = $\frac{9!}{(9-3)!}$

= $\frac{9!}{6!}$

= $\frac{9!}{6!}$

$$= 504$$

$$\therefore \text{required no. of ways} = 2 \times 504 \\ = 2016$$

one particular boy is always excluded?

→ Soln,

$$\text{Total no. of seats} = 4$$

$$\text{Total no. of boys} = 10$$

if one boy excluded, no of boys = 9

Now,

$$\text{no. of ways} = \frac{9!}{(9-4)!}$$

$$= \frac{9!}{5!}$$

$$= 3024$$

(ii) In how many ways can the letters of the word UNIVERSITY be arranged? How many such arrangements begin with 'U'? How many of these begin with U but do not end with Y?

Soln

Given: UNIVERSITY

$$\text{Total no. of letters} = 10 = n$$

$$'I' \text{ occurs } 2 \text{ times } (P) = 2$$

$$\text{no. of ways} = \frac{n!}{P!} = \frac{10!}{2!}$$

$$= 1814400$$

Arrangement begin with U

U, N, I, V, E, R, S, I, T, Y

required no. of arrangement = $\frac{9!}{2!}$

2!

$$= 181440$$

Arrangement begin with U and ends with Y

~~U~~ { N, I, V, E, R, S, T } Y

required no. of arrangement = $\frac{8!}{2!}$

2!

$$= 20160$$

Now,

Arrangement begins with U and ends with Y = $181440 - 20160$

$$= 161280$$

- (13) In how many ways can the letters of the word ELEMENT be arranged? How many of these arrangements are having all vowels together? Also, find the number of arrangements having no vowels together.

↳ Soln

Given: ELEMENT

No. of letters = 7 = (n)

'E' occurs 3 times (p) = 3

$$\text{No. of ways} = \frac{n!}{p!}$$

$$= \frac{7!}{3!}$$

$$3!$$

$$= 840$$

- Arrangement having All vowels together.

EEE, L, M, N, T

① ② ③ ④ ⑤

$$> \text{Block arrangement} = \frac{3!}{3!} = 1$$

$$> \text{Individual arrangement} = 5!$$

$$= 5!$$

$$= 120$$

∴ Required no. of arrangement having all vowels together = 120

Again

~~Not having 2 E's together~~

~~no. of arrangements not having 2 E's together~~

Arrangement having no vowels together.

① L ② E ③ M ④ E ⑤ N ⑥ E ⑦ T ⑧ E

3 E in 5 place can be arranged in $P(5,3)$

4 other letters in 4 place can be arranged in 4!

so,

$$\text{Required no. of arrangement} = \frac{P(5,3) \times 4!}{3!}$$

$$= \frac{60 \times 4 \times 3!}{3!}$$

$$= 240$$

∴ Required no. of arrangement having no vowels together = 240

①④ In how many ways can the letters of the word INTERNET be arranged so that each arrangement must begin and end with the same letter?

→ Soln,

No. of arrangement starts and ends with letter N

N { I, T, E, R, E, T } N

$$\text{No. of arrangement} = \frac{6!}{2! 2!} = 180$$

Similarly,

No. of arrangement starts and ends with letter T

T { I, N, E, R, N, E } T

$$\text{No. of arrangement} = \frac{6!}{2! 2!} = 180$$

No. of arrangement starts and ends with letters E

E { I, N, T, R, N, T } E

$$\text{No. of arrangement} = \frac{6!}{2! 2!} = 180$$

∴ Required no. of arrangement must begin and end with the same

letter = $180 + 180 + 180$

$$= 540$$

21.

⑮ In how many ways can a man post cards if 4 post boxes are available?

Soln Total post cards = 5

Total boxes = 4

1st post card has 4 ways

2nd post card has 4 ways

3rd post card has 4 ways

4th post card has 4 ways

5th post card has 4 ways

∴ Required no. of arrangement =

$$4 \times 4 \times 4 \times 4 \times 4$$

$$= 4^5$$

$$= 1024$$

2nd

• Circular permutation

→ It is an arrangement of objects in a circle.

Theorem:- The number of circular permutation of n distinct object is $(n-1)!$ if clockwise and anti-clockwise arrangements are different.

Note:- If clockwise and Anti-clockwise arrangement are same then the number of circular permutation is $\frac{(n-1)!}{2}$

[garland, bracelet, necklace].

- ⑩ In how many ways can 9 different colour beads be set in a bracelet.

→ Soln,

no. of colour (n) = 9

no. of ways for bracelet = $\frac{(n-1)!}{2}$

$$= \frac{(9-1)!}{2}$$

$$= \frac{8!}{2} = 20160 \text{ ways}$$

- ⑪ If 8 secretary level representatives of SAA-RC countries sit in round table conference, how many arrangement are possible? If Nepali and Indian secretary level sit always together, how many such arrangement will be there?

→ Soln,

No. of representatives (n) = 8

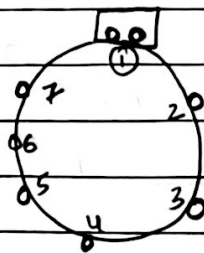
No. of ways = $(n-1)!$

$$= (8-1)!$$

$$= 5040$$

Again,

If nepali and Indian secretary level sit always together, now resulting number of representatives = $8-2+1 = 7$



For individual

No. of ways = $(n-1)!$

$$= (7-1)!$$

$$= 720$$

For together, Nepali and Indian,

No. of ways = 2!

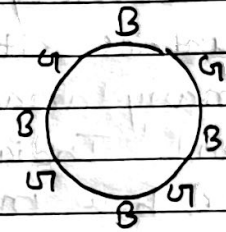
$$= 2$$

$\therefore$ required no. of ways = $720 \times 2 = 1440$

- ⑩ In how many ways can 4 boys and 4 girls sit alternatively on a round table?

→ Soln,

Because a round table has no fixed beginning or end we must fix the position of 1 person. (e.g. a boy) to break the circular symmetry.



With one boy fixed, there are 3 remaining boys who can be seated in 3 other 'Boy' seat. So,

arrangement of boys = $(4-1)! = 3!$ way

Arrangement of girls = $4!$ ways.

$\therefore$ Required number of ways = $3! \times 4!$

$$= 6 \times 24$$

$$= 144$$